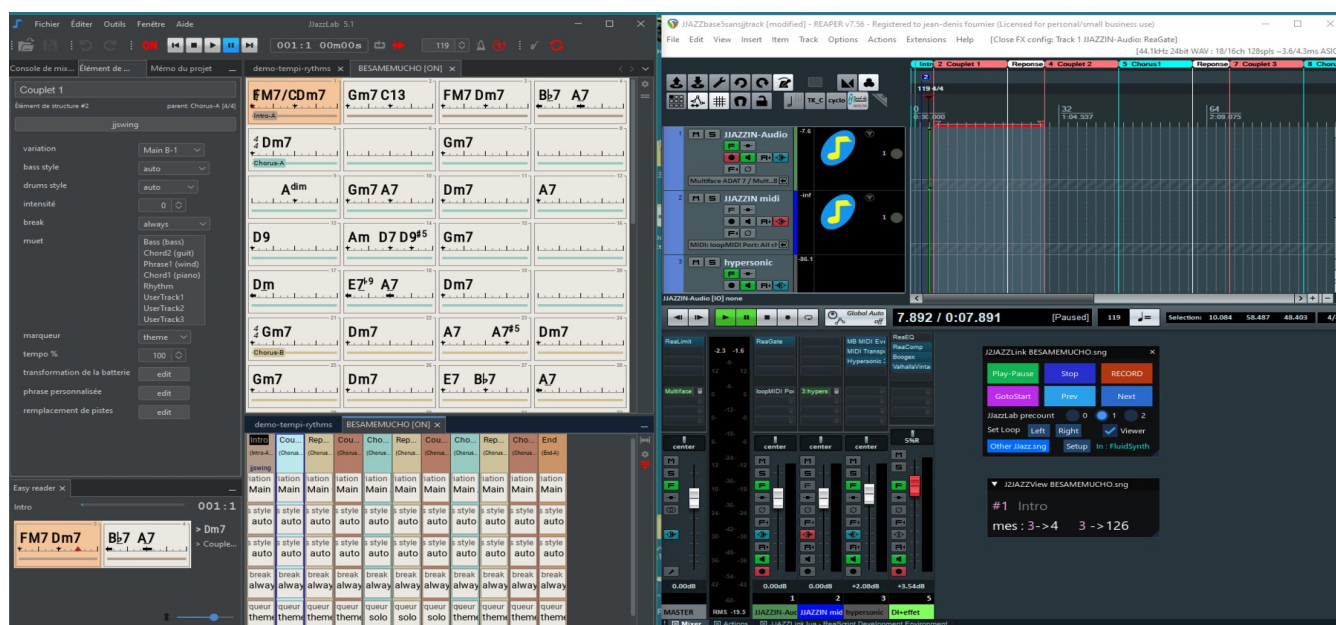
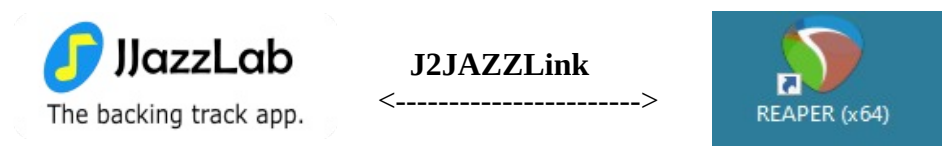


J2JAZZLink is a Reaper script written by Jean2 for use with JJazzLab and Reaper



On one hand, there's JJazzLab, a great arranger created by Jérôme Lelasseux, and on the other, there's Reaper. The two can work together thanks to a small script called J2JAZZLink.

With JJazzLab, you have access to an open-source arranger that lets you either create the song of the century with your own ideas or, more simply, practice your instrument by drawing from a massive library of standards. Plus, JJazzLab can import quite a few Band In A Box files, which is a huge plus. Add to that the fact that you can try it for free as much as you want—at the cost of a few seconds of pop-ups that are supposedly annoying but are actually kind of funny—and you'll see that JJazzLab is definitely worth checking out. Once you've seen all the “annoying” pop-ups, you end up making a donation, receive a code, and save a few seconds when opening JJazzLab. And you're happy, even if you miss the funny pop-ups... We're not going to review the Reaper DAW here, but just remind you that you can also try it for free before deciding to get a code for a minimal fee, given how good the program is. What's great about Reaper is that you can make it do almost anything you want by programming scripts and plugins.

J2JAZZLink is a script written in Lua within Reaper that allows you to extract the structure of a JJazzLab song as Reaper regions, and the chord names as markers in Reaper. You can then run Reaper while easily navigating the song's structure, with the chord “grid” displayed right in front of you. JJazzLab music can be produced

- *either offline (by importing audio or MIDI exported from JJazzLab into Reaper)
- *or in sync (by running JJazzLab and Reaper simultaneously with (fairly) good synchronization of the start times and tempo between the two applications). For this “live” use of JJazzLab, J2JAZZLink utilizes the ability to remotely control navigation within a song via MIDI notes, a feature offered by JJazzLab 5.2.

This allows you to use the Reaper environment to play with JJazzLab and, for example, record guitar or vocal parts “in sync” with JJazzLab.

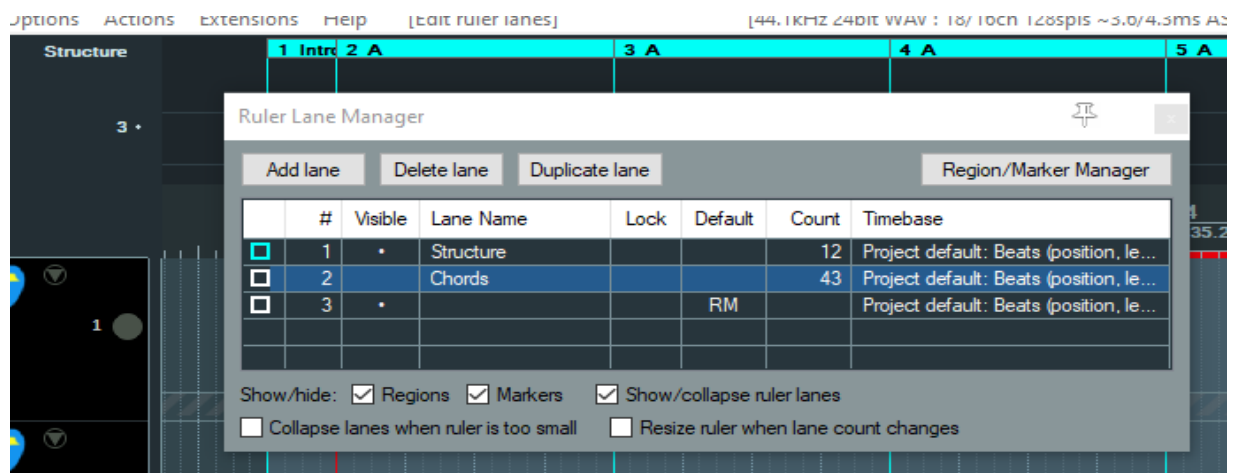
Installation

You need Reaper >= 7.72 and the ReaImGui library (see the Reaper website)

- 1) Place the J2JAZZLink.lua script in Reaper's scripts folder.
- 2) Place the J2JJlatencyspy.jsfx plugin in Reaper's effects folder (this plugin is used to estimate send latency when using JJazzLab "live")
- 3) To complete the installation, launch Reaper and activate J2JAZZLink from Reaper's Actions window (New Action, Load Reascript). It's very useful to assign this script to an icon in the toolbar (Customize toolbar).

Specifically, when J2JAZZLink loads a JJazzLab song, it integrates the following into Reaper:

- The song's structure, in the form of regions placed in ruler lane 1, taking into account the tempo, including any variations in speed and/or time signature, as allowed by JJazzLab. The region names come directly from the JJazzLab structure.
- Chord names are placed in Ruler Lane 2 as markers (names and positions only, no MIDI notes). This information is necessary to restore the environment when reopening a Reaper song based on a JJazzLab song; therefore, do not alter the contents of these two lanes. If you encounter any issues, you can always reload the JJazzLab song.



As mentioned above, it's helpful to hide Lane 2, this makes it easier to see the chords using the script's "grid" viewer.

If needed, you can use Lane 3 as a ruler for markers or custom regions; this shouldn't interfere with the script.

Utilisation

Control Panel

As you might expect, the "Other Jjazz song" button opens a JJazzLab song.
Note: Some older songs won't open at all, or may be with missing chords in the chord chart. For example, those from the Real Book. This is likely due to JJazzLab version



compatibility issues. Solution: Re-save them after making a fake edit in JJazzLab 5.2 (e.g., in JJazzLab, select all chords, transpose+, transpose-, save); after that, it should work.

To get the correct bar numbers, remember to set the start bar to -2 in Reaper's "Project Settings."

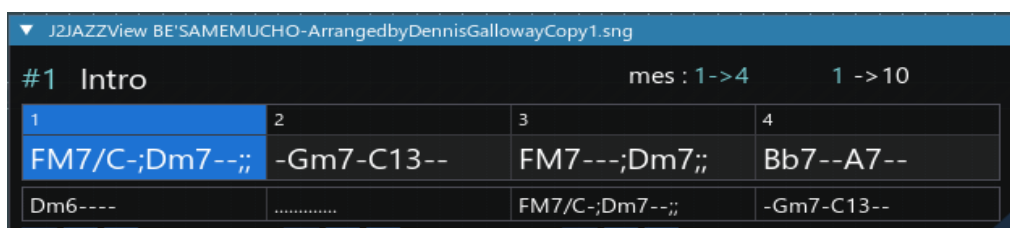
The "Set Loop" buttons let you define loops aligned with key points in the song's structure.

The Chord Grid Viewer

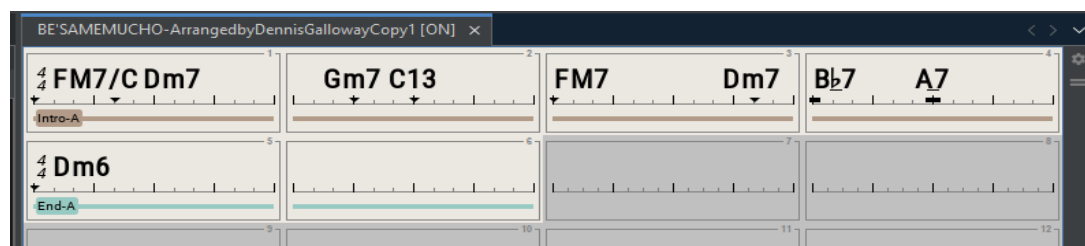
In grid view, the blue cursor moves, and when it reaches the end of the line, the next line (shown in small print below) moves up one position.

At the bottom of the window, you can set the number of measures per line, the position of measure 1 on the line, the font size, the number of future lines, and whether or not to enable a slightly rhythmic display in the grid, based on a rudimentary coding of durations:

- 1 beat . 1/2 beat ; 1/3 beat , 1/4 beat



Then hide the settings by rolling up the bottom of the window.



The Original JJazzLab

Working with JJazzLab Offline

In Setup 4), keep the "Force JJazzLab OFF" checkbox checked.

It's best to start by loading the JJazzLab song with the script before importing the media (MP3 or MIDI); this resolves the tempo issue in Reaper and indicates where to place the import.

The main benefit of the script in this case is the ability to navigate the song's structure and view the chord progression while playing along.

Working with JJazzLab "live"

This involves running JJazzLab and Reaper simultaneously, with the start times and tempo of both applications (fairly) well synchronized. The benefit is being able to edit JJazzLab in real time.

In this setup, JJazzLab receives remote MIDI commands sent from Reaper via a MIDI loopback program such as "Loop MIDI." This software allows you to define MIDI ports between applications. You'll need a port that acts as an output on the Reaper side and as an input on the JJazzLab side.

To receive the signal from JJazzLab in Reaper, you must have set up a track dedicated to receiving the signal from JJazzLab and outputting the music (either an audio signal from FluidSynth or MIDI). See below for additional information.

Open JJazzLab, start a song, and verify that you are receiving the music in Reaper.

Open J2JAZZLink. The first time you use it, go through the setup (see below) to configure items 1), 4), and 5). Your settings will be saved for future sessions of J2JAZZLink.

Load the same song into J2JAZZLink (using the “Other Jjazz sng” button).

You can, of course, load a different song whenever you want; the regions will simply be replaced. This is great for working on standards on the fly with your guitar, piano, or other instrument.

To set up a loop, simply define it in Reaper using the Set Loop buttons—which set the start point (left or right) based on the region where the cursor is located—and select the corresponding structural elements in JJazzLab. If you want to loop the entire song, you don’t need to make a selection in JJazzLab.

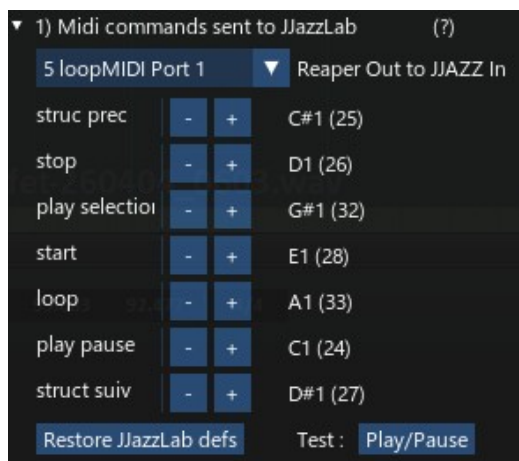
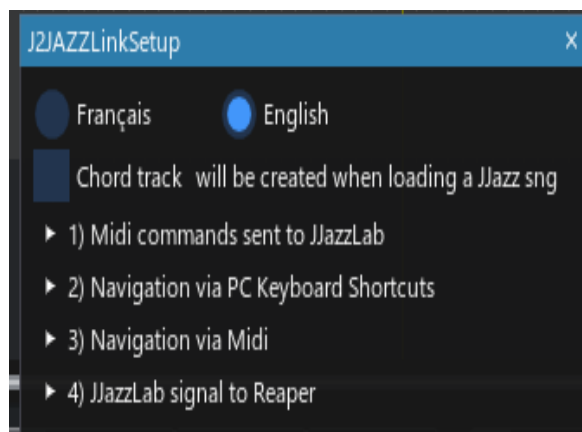
Important notes:

- The JJazzLab count must be set to Off for navigation to work properly. Reaper’s metronome will take its place.
- If Reaper and JJazzLab are no longer in sync (with a “live” signal from JJazzLab), pressing Stop should resolve the issue.
- The tempo base (bpm basis) is currently not accessible via script in Reaper. Upon import, it is assumed to be the quarter note (1/4). If there are measures in 6/8 or 12/8 time, the tempo will be multiplied by 1.5 to match JJazzLab. You can correct this manually afterward by setting the correct tempo and a bpm basis of a dotted quarter note on the tempo marker.

Setup

Such a technical product (just kidding) wouldn't work without a few adjustments: the Setup.

It covers almost everything, with helpful tips...



1) MIDI Commands to JJazzLab (for “live” use)

Specify here the MIDI port that will send commands from Reaper to JJazzLab. By default, these are the same commands as those in JJazzLab; it's best to leave them as is.

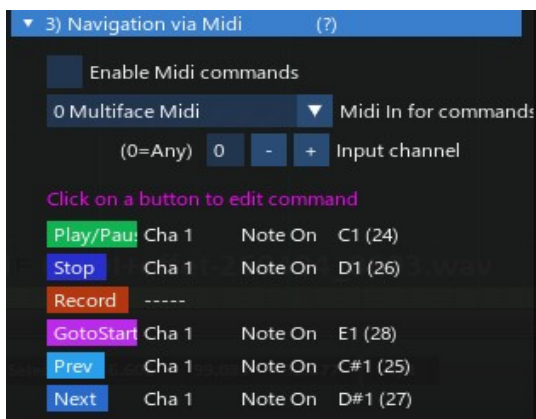
With this Play/Pause button, you can do a quick test to see if JJazzLab is receiving the signal properly.

2) Navigation keyboard shortcuts, which you can redefine (using only “single” keys, no CTRL or Shift)



Note: Keyboard shortcuts only work if the script has focus; otherwise, Reaper’s standard shortcuts will work.

To return focus to the script, you can click anywhere at the top of the viewer, where nothing is editable (so you don’t have to aim too precisely).



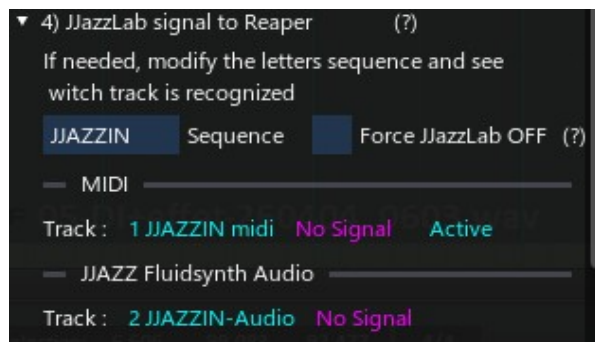
3) Controlling the script via MIDI

You can control the script from a MIDI keyboard; it should work, but it hasn’t been tested much since it’s not very useful when you’re sitting in front of the computer.

(Disabled here)

4) Signal from JJazzLab to Reaper (for “live” use)

Here, you can verify whether the script has correctly identified the tracks through which the “live” audio signal from JJazzLab is fed into Reaper. To do this, it searches for the first tracks that contain the (customizable) character sequence JJAZZIN and have a stereo audio or MIDI input.



For example, above are the two channels in my template; I use one or the other depending on the style of the track (FluidSynth’s audio is perfect for jazz).

If no live input channels are defined, the script assumes that you are working with files (MP3 or MIDI) from JJazzLab in Reaper channels, and therefore that JJazzLab is OFF.

If live input tracks are defined, the script will wait for a signal to identify its type and adjust the initial latency accordingly, unless you check “Force JJazzLab OFF.”

5) Latency Compensation (for “live” use)

This section appears only if “live” input channels are present.

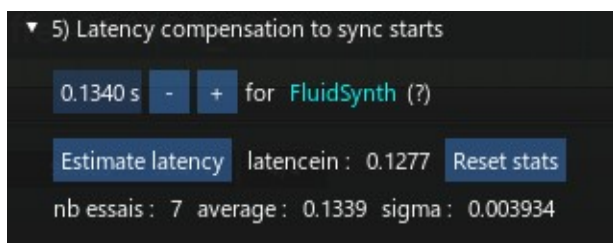
Depending on the signal you’re using, you’ll find the settings here for MIDI latency or FluidSynth latency. These settings determine the synchronization of playback with JJazzLab’s “live” inputs.

The values correspond to the time between Reaper sending the Play command and JJazzLab outputting the music. There is a small tool that lets you estimate these two values using the “Estimate latency” button.

To do this, start the JJazzLab countdown to ensure you receive a steady click.

Notes:

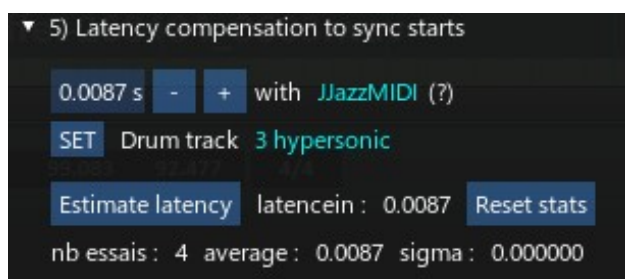
- After a change (on → off or off → on) to the JJazzLab countdown, latency is often higher during JJazzLab’s first subsequent playback. It returns to “normal” afterward, so it’s best to perform a play/stop sequence after such a change to the countdown to get an accurate measurement of latency.
- To measure latency, a send is created from the channel playing the click to a mysterious track called “JJstarter” (see “How are starts triggered?” below). You can safely delete this send once the latency has been measured. Normally, it disappears on its own when a JJazzLab song is loaded into the script.



With an audio signal, latency fluctuates quite a bit, and you can get an average value by clicking the “Estimate latency” button several times.

As shown here, over 7 trials, the average is 0.1339 s, with a standard deviation of about 4 ms. The retained value is 0.1340

This might seem concerning: if you’re recording a guitar, the backing track from JJazzLab will never be exactly in the same place when you hit play! But in practice, these few milliseconds aren’t too much of a problem—unless, perhaps, you’re planning to record an album. When working on the instrument, this tool remains perfectly usable, and you can always import an audio file and work with JJazzLab offline if you want to avoid these fluctuations.



With a MIDI signal routed to VSTIs, the window is slightly different: you also need to specify which drum track will provide the click in Reaper (select that track and click “SET”)

Note that the latency is much lower—and also more consistent—than with the audio signal. On my system, it’s almost always exactly 0.0087 seconds.

Additional Information

1) MIDI communication between JJazzLab and Reaper : MIDI loopback (for “live” use)

MIDI communication between Reaper and JJazzLab is handled using MIDI loopback software, such as LoopMIDI. This software sets up as many ports as you want, which are recognized by Reaper and JJazzLab as hardware ports. Personally, I use 2 ports:



Name	Total data	Throughput / sec.
loopMIDI Port	91965	0 Byte
loopMIDI Port 1	507	0 Byte

It is also this JJStarter track that contains the J2JJlatencyspy.jsfx plug-in, which is used to estimate the offset for synchronization.

You can, of course, make this track visible using Reaper's track manager.